GEETHANJALI INSTITUTE OF SCIENCE & TECHNOLOGY (AUTONOMOUS)

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QUESTIONBANK(DESCRIPTIVE)

Subject Name with Code: Building Materials and Services & 23A0152T

Course & Branch: B.Tech & CSE,CS,DS,AIML,ECE

Year& Semester: IV & I **Regulation: RG23**

UNIT – I STONES AND BRICKS, TILES

S.No.	Question	[BT Level] [CO][Marks]
	2 Marks Questions (Short)	
1.	Define building stones and classify them based on geological origin.	L1 CO1 2M
2.	What is quarrying? List the methods of quarrying stones.	L2 CO1 2M
3.	Define dressing of stones. Mention its types.	L2 CO1 2M
4.	List the constituents of brick earth and state their functions.	L2 CO1 2M
5.	Define fly ash and distinguish between Class F and Class C fly ash.	L2 CO1 2M
6.	What are ceramics? Mention any four properties of ceramics.	L1 CO1 2M
7.	Define seasoning of timber and list its methods.	L2 CO1 2M
8.	State the defects in timber.	L2 CO1 2M
9.	List the different alternate materials used instead of timber.	L2 CO1 2M
10.	What are the characteristics of good paint and good plastic?	L2 CO1 2M
	Descriptive Questions (Long)	
11.	Explain the classification of building stones, methods of quarrying, structural requirements and dressing of stones with neat sketches.	L1 CO1 10M
12.	Describe the composition of brick earth, manufacture of bricks and structural requirements of good bricks.	L2 CO1 10M
13.	Explain fly ash in detail, including its source, classification, properties, chemical composition, pozzolanic properties, advantages, disadvantages and applications in construction.	L3 CO1 10M
14.	Explain ceramics with reference to raw materials, manufacturing process, properties, types, advantages, disadvantages and uses in construction.	L2 CO1 10M
15.	Describe the structure, types, properties, seasoning methods, defects and characteristics of good timber.	L1 CO1 10M
16.	Discuss the alternate materials for timber. Compare steel, aluminium, plastics, FRP, plywood, MDF, glass and engineered bamboo with timber.	L3 CO1 10M
17.	Explain aluminium as a building material. Discuss its source, properties, manufacturing process, alloys, advantages, disadvantages and applications in construction.	L2 CO1 10M
18.	Explain glass as a building material. Describe its raw materials, manufacturing process, properties, types, advantages, disadvantages and uses in construction.	L2 CO1 10M
19.	Explain paints and plastics used in construction. Discuss their composition, types, properties, applications, advantages and disadvantages.	L3 CO1 10M
20.	Write short notes on the following building materials: (a) Galvanized Iron (GI) (b) Fibre Reinforced Plastic (FRP) (c) Glass Bricks (d) Steel.	L2 CO1 10M

UNIT – II CEMENT &ADMIXTURES

S.No.	Question	[BT Level]

		[CO][Marks]
2 Marks Questions (Short)		
1.	Define cement and state its functions.	L1 CO2 2M
2.	List the different types of cement.	L2 CO2 2M
3.	Name the main ingredients of cement and state their functions.	L2 CO2 2M
4.	What is hydration of cement?	L1 CO2 2M
5.	Mention any four field tests of cement.	L2 CO2 2M
6.	Define standard consistency of cement.	L1 CO2 2M
7.	Differentiate between initial setting time and final setting time.	L2 CO2 2M
8.	What is soundness of cement? Mention the methods used to determine it.	L2 CO2 2M
9.	Define admixtures. Classify admixtures.	L2 CO2 2M
10.	Distinguish between mineral admixtures and chemical admixtures.	L2 CO2 2M
Descriptive Questions (Long)		
11.	Explain the different types of cement, their properties, advantages and applications in construction.	L1 CO2 10M
12.	Describe the ingredients, manufacture and chemical composition of Ordinary Portland Cement with a neat flow chart.	L2 CO2 10M
13.	Explain the hydration of cement. Discuss the hydration products, heat of hydration, factors affecting hydration and its importance.	L3 CO2 10M
14.	Explain the field and laboratory tests conducted on cement. Discuss their purpose and relevant IS codes.	L2 CO2 10M
15.	Describe the fineness test of cement by Sieve Method and Blaine Air Permeability Method with principles, procedure and significance.	L2 CO2 10M
16.	Explain the Standard Consistency Test, Initial and Final Setting Time Test, and Soundness Test of cement with neat sketches of the Vicat Apparatus and Le-Chatelier Apparatus.	L3 CO2 10M
17.	Define admixtures. Explain the different types of mineral admixtures and their properties, advantages and applications in concrete.	L1 CO2 10M
18.	Explain the different types of chemical admixtures. Discuss their functions, advantages, disadvantages and applications in concrete.	L2 CO2 10M
19.	Compare mineral admixtures and chemical admixtures. Explain their characteristics, functions and uses with suitable examples.	L3 CO2 10M
20.	Explain the advantages, disadvantages and engineering applications of admixtures in improving the properties of concrete.	L2 CO2 10M

UNIT – III BUILDING COMPONENTS

S.No.	Question	[BT Level] [CO][Marks]
2 Marks Questions (Short)		
1.	Define a lintel and state its functions.	L1 CO3 2M
2.	Define an arch and list its main components.	L2 CO3 2M
3.	Classify walls based on their structural functions.	L2 CO3 2M
4.	What is a vault? Mention its types.	L2 CO3 2M
5.	List the components of a staircase.	L2 CO3 2M
6.	State the requirements of a good floor.	L2 CO3 2M
7.	Classify roofs based on their shape and construction.	L2 CO3 2M
8.	Define a foundation and classify the types of foundations.	L2 CO3 2M
9.	What is a Damp Proof Course (DPC)? Mention its importance.	L2 CO3 2M
10.	Define joinery. List the common types of doors and windows.	L2 CO3 2M
Descriptive Questions (Long)		
11.	Explain the different types of lintels. Discuss their advantages, disadvantages and applications in building construction.	L1 CO3 10M
12.	Explain the classification of arches. Describe the components, advantages,	L2 CO3 10M

	disadvantages, failures and applications of arches with neat sketches.	
13.	Describe the classification of walls. Explain the construction, advantages and applications of load-bearing walls, cavity walls, shear walls and retaining walls.	**L3 CO3 10M**
14.	Explain the different types of vaults. Discuss their components, advantages, disadvantages and engineering applications with neat sketches.	**L2 CO3 10M**
15.	Explain the components, standard dimensions, types and requirements of a good staircase. Discuss the advantages and disadvantages of different staircase systems.	**L2 CO3 10M**
16.	Classify the different types of floors and roofs. Explain their construction, advantages, disadvantages and applications in building construction.	**L3 CO3 10M**
17.	Explain the classification of foundations. Describe shallow and deep foundations with neat sketches, advantages and applications.	**L2 CO3 10M**
18.	Explain Damp Proof Course (DPC). Discuss the causes and effects of dampness, types of DPC, damp-proofing methods, materials used and modern waterproofing systems.	**L3 CO3 10M**
19.	Explain joinery in buildings. Describe the different types of doors and windows, their components, materials, selection criteria and applications.	**L2 CO3 10M**
20.	Write short notes on the following building components: (a) Modern wall systems (b) Modern staircase systems (c) Modern roofing systems (d) Modern joinery systems.	**L2 CO3 10M**

UNIT – IV MORTARS, MASONRY AND FINISHING MORTARS

S.No.	**Question**	**[BT Level] [CO][Marks]**
	2 Marks Questions (Short)	
1.	Define lime mortar and cement mortar. State their applications.	**L1 CO4 2M**
2.	What is brick masonry? Mention the common types of brick bonds.	**L2 CO4 2M**
3.	Classify stone masonry with suitable examples.	**L2 CO4 2M**
4.	Define composite masonry. Mention its advantages.	**L2 CO4 2M**
5.	What is plastering? List the common defects in plastering.	**L2 CO4 2M**
6.	Define cladding and ACP (Aluminium Composite Panel).	**L1 CO4 2M**
7.	What is formwork? State the requirements of good formwork.	**L2 CO4 2M**
8.	Define scaffolding and list its different types.	**L2 CO4 2M**
9.	What is shoring? Name the different types of shoring.	**L2 CO4 2M**
10.	Define underpinning. Mention the situations where underpinning is required.	**L2 CO4 2M**
	Descriptive Questions (Long)	
11.	Explain lime mortar and cement mortar. Compare their composition, properties, advantages, disadvantages and applications.	**L1 CO4 10M**
12.	Explain brick masonry in detail. Discuss the types of brick masonry, brick bonds, requirements of good brick masonry, advantages and applications with neat sketches.	**L2 CO4 10M**
13.	Describe stone masonry and composite masonry. Explain their types, construction methods, advantages, disadvantages and applications.	**L3 CO4 10M**
14.	Explain concrete masonry and reinforced brick masonry. Discuss their construction procedure, advantages and engineering applications.	**L2 CO4 10M**
15.	Explain finishing works in buildings. Discuss plastering, pointing, painting, cladding, tiles and Aluminium Composite Panels (ACP) with their advantages and applications.	**L3 CO4 10M**
16.	Explain formwork in detail. Discuss its components, requirements, types, standards, design considerations and removal procedures.	**L2 CO4 10M**
17.	Describe scaffolding. Explain its components, types, design considerations, safety precautions and applications in construction.	**L2 CO4 10M**
18.	Explain shoring in detail. Discuss the different types of shoring, their	**L3 CO4 10M**

	applications, safety measures and advantages in construction.	
19.	Explain underpinning. Discuss the need, causes, methods, advantages, limitations and applications of underpinning in building construction.	**L2 CO4 10M**
20.	Write short notes on the following: (a) ACP (Aluminium Composite Panel) (b) Formwork Standards (c) Design Considerations for Scaffolding (d) Methods of Underpinning.	**L2 CO4 10M**

UNIT – V BUILDING SERVICES

S.No.	Question	[BT Level] [CO][Marks]
	2 Marks Questions (Short)	
1.	Define plumbing services and state their objectives.	**L1 CO5 2M**
2.	Explain the direct and indirect water distribution systems.	**L2 CO5 2M**
3.	Define sanitary plumbing. Mention the different types of sanitary fittings and traps.	**L2 CO5 2M**
4.	List the common pipe materials used in plumbing and state their applications.	**L2 CO5 2M**
5.	Define ventilation. State the functional requirements of good ventilation.	**L2 CO5 2M**
6.	Classify the different systems of ventilation.	**L2 CO5 2M**
7.	What is air conditioning? List its essential components.	**L2 CO5 2M**
8.	Define acoustics. State the characteristics of sound.	**L2 CO5 2M**
9.	Classify fires based on fuel source (Class A, B, C, D and K/F).	**L2 CO5 2M**
10.	What are fire-resistant materials? Name any four fire-fighting systems used in buildings.	**L2 CO5 2M**
	Descriptive Questions (Long)	
11.	Explain plumbing services in buildings. Discuss the water distribution system, sanitary plumbing, fittings, pipe materials and plumbing maintenance.	**L1 CO5 10M**
12.	Explain ventilation in detail. Discuss its functional requirements, systems of ventilation, factors affecting ventilation, advantages and applications.	**L2 CO5 10M**
13.	Describe air conditioning. Explain its essentials, components, different types, advantages, limitations and applications in buildings.	**L3 CO5 10M**
14.	Explain acoustics in building design. Discuss the characteristics of sound, sound absorption, acoustic materials and the principles of acoustic design.	**L2 CO5 10M**
15.	Describe the acoustic design requirements for auditoriums, classrooms and cinema halls. Explain the importance of sound insulation and reverberation control.	**L3 CO5 10M**
16.	Explain fire protection in buildings. Discuss the objectives, importance, fire hazards, fire triangle, fire tetrahedron and methods of fire spread.	**L2 CO5 10M**
17.	Classify fires based on fuel source. Explain the fire-resistant materials, fire-resistant construction and various fire-fighting systems used in buildings.	**L2 CO5 10M**
18.	Explain the fire safety measures adopted in buildings. Discuss emergency exits, evacuation plans, fire alarms, detectors, sprinklers and maintenance of fire protection systems.	**L3 CO5 10M**
19.	Compare natural, mechanical and hybrid ventilation systems. Discuss their working principles, advantages, disadvantages and applications.	**L3 CO5 10M**
20.	Write short notes on the following: (a) Water Distribution System (b) Pipe Materials (c) Air Conditioning Systems (d) Fire-Fighting Equipment.	**L2 CO5 10M**

Signature of the Staff:

Signature of Department Academic Committee Member 1:

Signature of Department Academic Committee Member 2:

Signature of Department Academic Committee Member 3: